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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H2 BIOLOGY

9648/03

Paper 3 Applications Paper and SPA Planning Task

23 September 2016

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A:

Consists of 3 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 1 SPA Planning Task

Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Section C:

Consists of 1 Free-Response Question.

Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Sections A, B and C are to be submitted **separately**.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

For Examiner's Use	
Section A [40]	
1	/ 13
2	/ 12
3	/ 15
Section B[12]	
Section C [20]	
Total [72]	

This document consists of **16** printed pages.

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Section A: Structured Questions (40 marks)

Answer **all** questions in this section.

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Question 1

- (a) **Fig. 1.1** shows two types of plasmids, pAMP and pKAN. pAMP carries an ampicillin-resistant (amp^r) gene and pKAN carries a kanamycin-resistant (kan^r) gene. Restriction sites for BamHI and HindIII are found in both plasmids.

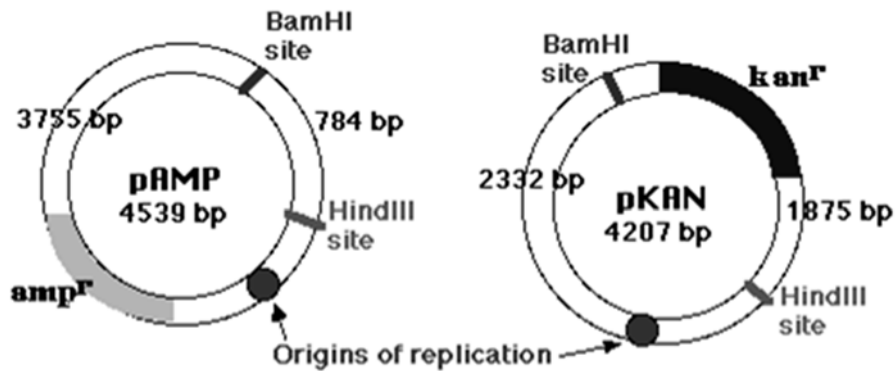


Fig. 1.1

A ligation solution of these two plasmids is mixed together with the enzymes BamHI and HindIII. The pAMP and pKAN fragments were then allowed to anneal, resulting in the plasmid shown in **Fig. 1.2**.

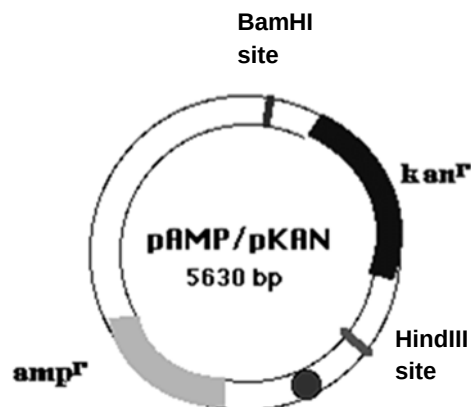


Fig. 1.2

- (i) In addition to the recombinant plasmid shown in **Fig. 1.2**, three other recombinant plasmids are also formed. Draw and label clearly these three recombinant plasmids in the space provided below. [3]

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- (ii) *E. coli* is mixed with the ligation solution in (a). Describe and explain how *E. coli* carrying plasmid shown in **Fig. 1.2** can be selected. [2]

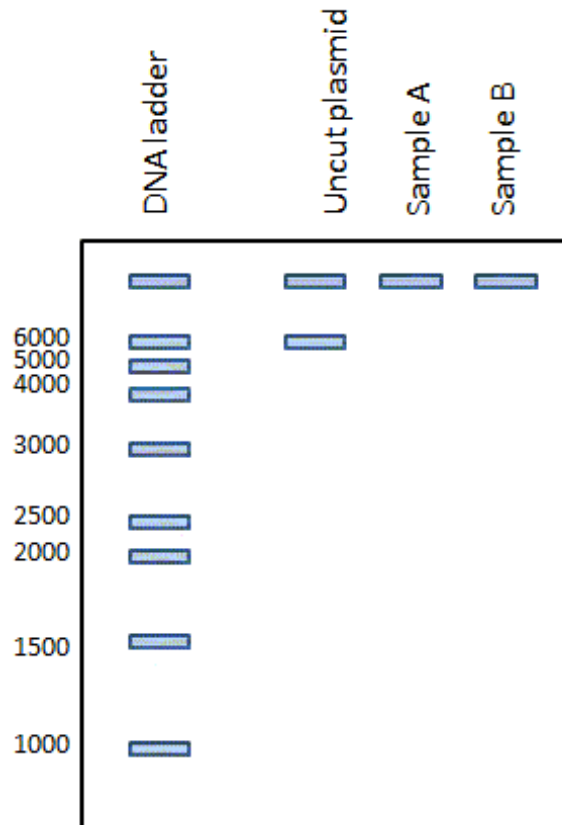
- (iii) With reference to **Fig. 1.2**, in addition to the restriction sites for BamHI and HindIII, a single restriction site for EcoR1 was found within the sequence of the *kan^r* gene. To clone a human gene with a molecular size of 600 bp into bacterial cells, EcoR1 was used to cut the human gene and the plasmid in **Fig. 1.2**. The cut human gene and plasmid are then mixed together with DNA ligase. The ligated DNA mixture is then used for the transformation of *E. coli*. The bacteria is then grown on a nutrient agar plate.

To identify *E. coli* cells that have taken up the recombinant plasmid with the human DNA, a scientist isolates two types of plasmids from 2 bacteria colonies respectively. He then subjects the two DNA samples to restriction digestion using enzymes BamHI and HindIII, producing DNA fragments of the following sizes.

Sample	Fragments (kb)
A	3755, 1875
B	3755, 2475

Suggest which sample of transformed bacterial cells is able to survive on an agar plate containing ampicillin but **NOT** on an agar plate containing kanamycin. Explain your answer. [3]

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(b) DNA of organisms may be stored in genomic DNA libraries or cDNA libraries.

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use*

- (i)** Outline the roles of one enzyme used in the formation of the cDNA in a cDNA library. [1]

- (ii)** Explain why cDNA libraries made from the same type of cell at different times in the life of the cell may vary, whilst a genomic library for that organism will always be the same. [3]

Total: [13]

(a) Although the DNA from different individuals is more alike than different, there are many chromosomal regions that exhibit a great deal of diversity. Such variable sequences are termed polymorphic and provide the basis for disease diagnosis, forensic identification and paternity testing.

(i) Many repeat polymorphisms are highly polymorphic, having tens of different alleles. Repeat polymorphisms also exhibit high level of heterozygosity. Explain what you understand by the term “heterozygosity”. [1]

- (ii)** Repeat polymorphisms used in forensic biology are neutral mutations, which do not affect protein functions. Suggest one location where such mutations are primarily found. [1]

- (b) An experiment examines a VNTR on human chromosome 1 known as D1S80. Each repeat unit in this VNTR is 16 base-pairs (bp) long. Most individuals have between 14 and 40 copies of the repeat at the D1S80 locus.

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In this experiment, polymerase chain reaction (PCR) is used to determine the number of repeated DNA sequences at the D1S80 locus. Each DNA sample is obtained from a single swab of cheek cells from a volunteer. Primers (see arrows) were used for the PCR as shown in **Fig. 2.1**.

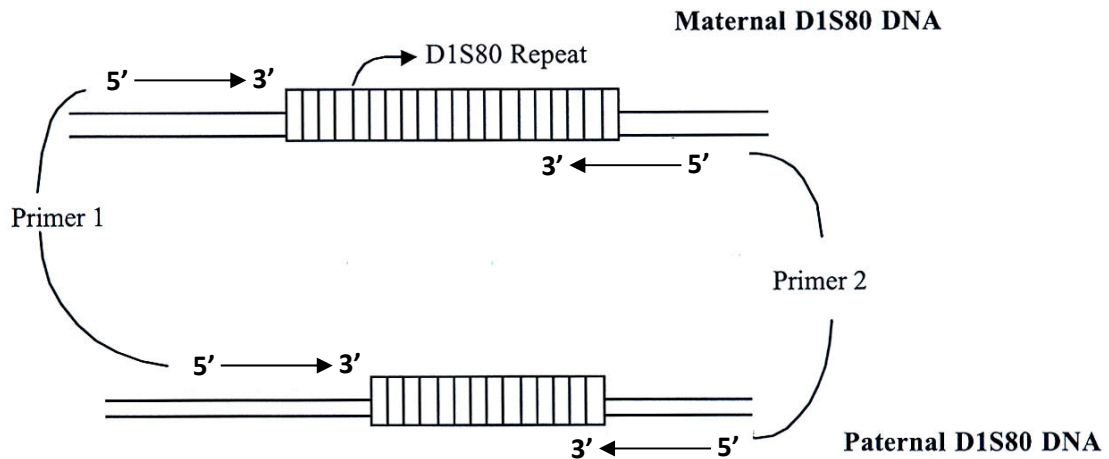


Fig. 2.1

Note:

Primer 1 : 5'-GAA ACT GGC CTC CAA-3' (15-mer)

Primer 2: 5'-GTC TTG TTG GAG ATG-3' (15-mer)

To compare the genotypes from a number of volunteers, aliquots of the respective PCR products and a DNA molecular marker are loaded onto the wells of a DNA chip. Microfluidic electrophoresis is carried out. Following computer analysis, the PCR products of one of the volunteers appear as distinct peaks with assigned base-pair sizes on an electropherogram, as shown in **Fig. 2.2**. The sizes of the PCR products can then be used to determine the specific number of repeats within each VNTR allele.

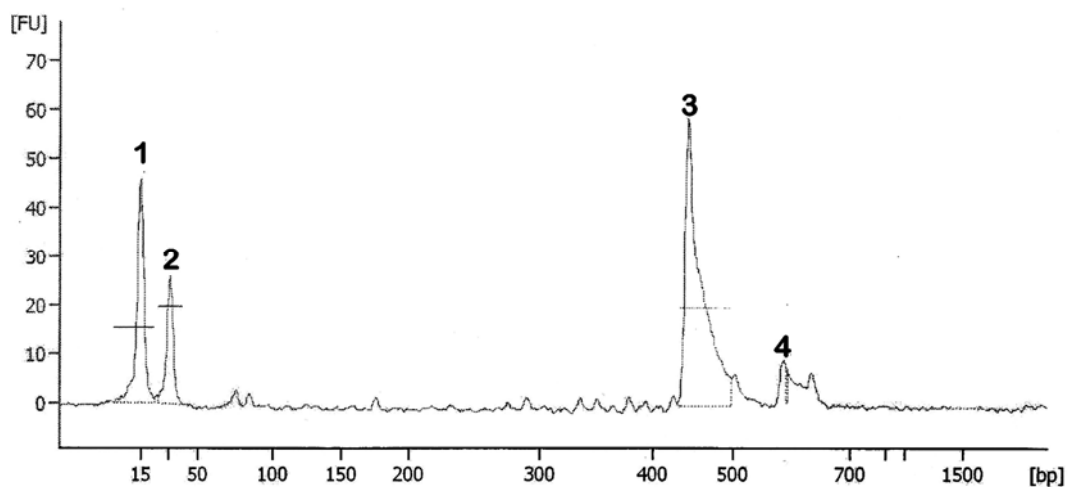


Fig. 2.2

The size of DNA corresponding to each of the peak is shown in **Table 2.1**.

Table 2.1

Peak	Size / bp
1	15
2	30
3	443
4	590

- (i) Explain why peaks **1** and **2** appear in the electropherogram. [2]

- (ii) Based on the DNA sizes corresponding to peaks **3** and **4**, calculate the number of repeats present in each VNTR allele of the volunteer. [3]

- (iii) Suggest why there is an anomaly in the number of repeats found in one of the VNTRs, besides instrumental error. [2]

- (iv) Discuss the benefits of using PCR analysis in place of RFLP analysis for this experiment. [3]

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Total: [12]

Question 3

- (a) The first gene therapy trial for an inherited disorder was initiated on 14 September 1990. The patient, Ashanti DeSilva, suffered from a very rare recessively inherited disorder, adenosine deaminase (ADA) deficiency.

An inherited deficiency of ADA has particularly severe consequences in the case of T lymphocytes. As a result, ADA-deficient patients suffer from severe combined immunodeficiency (SCID).

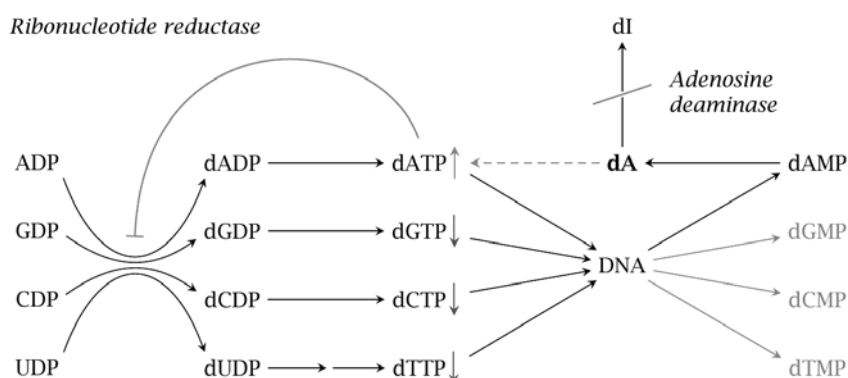


Fig. 3.1

- (i) Using the information provided, explain how ADA deficiency can lead to SCID. [3]

The novel ADA gene therapy approach conducted on Ashanti DeSilva involved the following steps:

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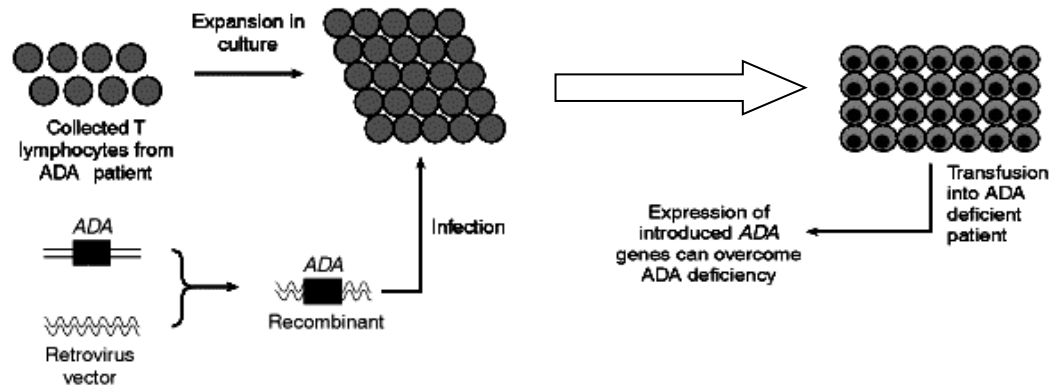


Fig. 3.2

The protocol was reviewed a dozen times by seven regulatory committees before it was finally approved by the RAC in July 1990, and by the FDA two months later. The gene therapy was considered to be a success for Ashanti but the response was far more limited in the second patient.

- (ii) Suggest an improvement to the procedure shown in **Fig. 3.2** that will increase the success rate of the treatment. Explain your answer. [2]

- (b) (i) ADA-SCID, cystic fibrosis and sickle cell anaemia are examples of autosomal recessive disorders. Such disorders are often prime candidates to be treated by gene therapy.

Explain why it is easier to perform gene therapy when a mutant allele is recessive instead of dominant. [2]

- (ii) Although the use of gene therapy for treatment of genetic disorders in humans seems promising, there are still many concerns to be addressed.

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Explain the factors that may keep gene therapy from becoming an effective treatment for genetic disorders. [3]

- (c) Scientists have come up with an alternative method of generating pluripotent cells, which is to genetically reprogramme adult cells to an embryonic stem cell-like state. Genetic reprogramming is carried out by using deactivated retroviruses to introduce the genes of four transcription factors into adult cells from a patient (**Fig. 3.3**). The reprogrammed cells, called induced pluripotent stem (iPS) cells, are specific to the patient from which the adult cells were taken

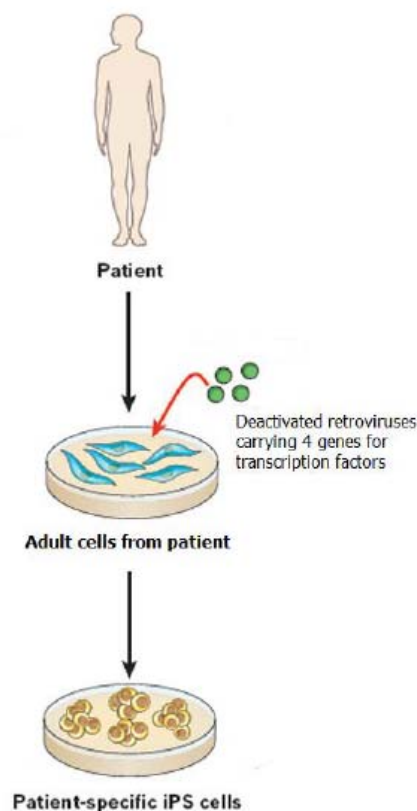


Fig. 3.3

- (i) State two advantages of using iPS cells instead of embryonic stem cells for research and clinical trials. [2]

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use*

- (ii) Suggest how the expression of such a small number of transcription factors in adult cells could genetically reprogramme these adult cells to an embryonic stem cell-like state. [3]

Total: [15]

Section B: SPA Planning Task (12 marks)

Write your answers on the writing paper provided.

A **NIL RETURN** is required.

Question 4

Beetroots are plants that have storage roots that are 5 to 10cm in diameter. The storage tissues of these plants have cells that contain betacyanin (red pigment) in the cell vacuole. The betacyanin pigment cannot pass through membranes, but can pass through the cellulose cell walls if the membrane integrity is disrupted.

Physical damage to the storage roots of beetroot, for example by cutting, causes large loss of pigment.

Using this information and your own knowledge, design an experiment to determine the effect of temperature on beetroot membrane integrity.

You must use:

- Beetroot
- Distilled water

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, syringes, glass rods etc.
- White card
- White tile
- Knife, scapel, cork borers
- Ruler
- Blunt forceps
- Stopwatch
- Thermometer
- Access to a kettle to boil water and ice
- Marker pen
- 5% betacyanin
- colorimeter

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible;
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

Total: [12]

Section C: Free-Response Question (20 marks)

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 5

- (a)** Biofuels can be obtained from the conversion of cellulosic biomass, which is both abundant and renewable. However, the enzymes and pretreatment processes involved are very expensive. One approach is to genetically engineer plants to produce cellulase so as to enhance the conversion of cellulose into fermentable sugars and reduce the need for pretreatment processes.

Describe how one can mass produce more plants that has already been genetically modified to synthesize cellulase. [7]

- (b)** Explain the significance of genetic engineering in GM salmon and Bt corn in solving the demand for food in the world. [7]

- (c)** Discuss the social and ethical implications of Bt corn. [6]

Total: [20]

END OF PAPER